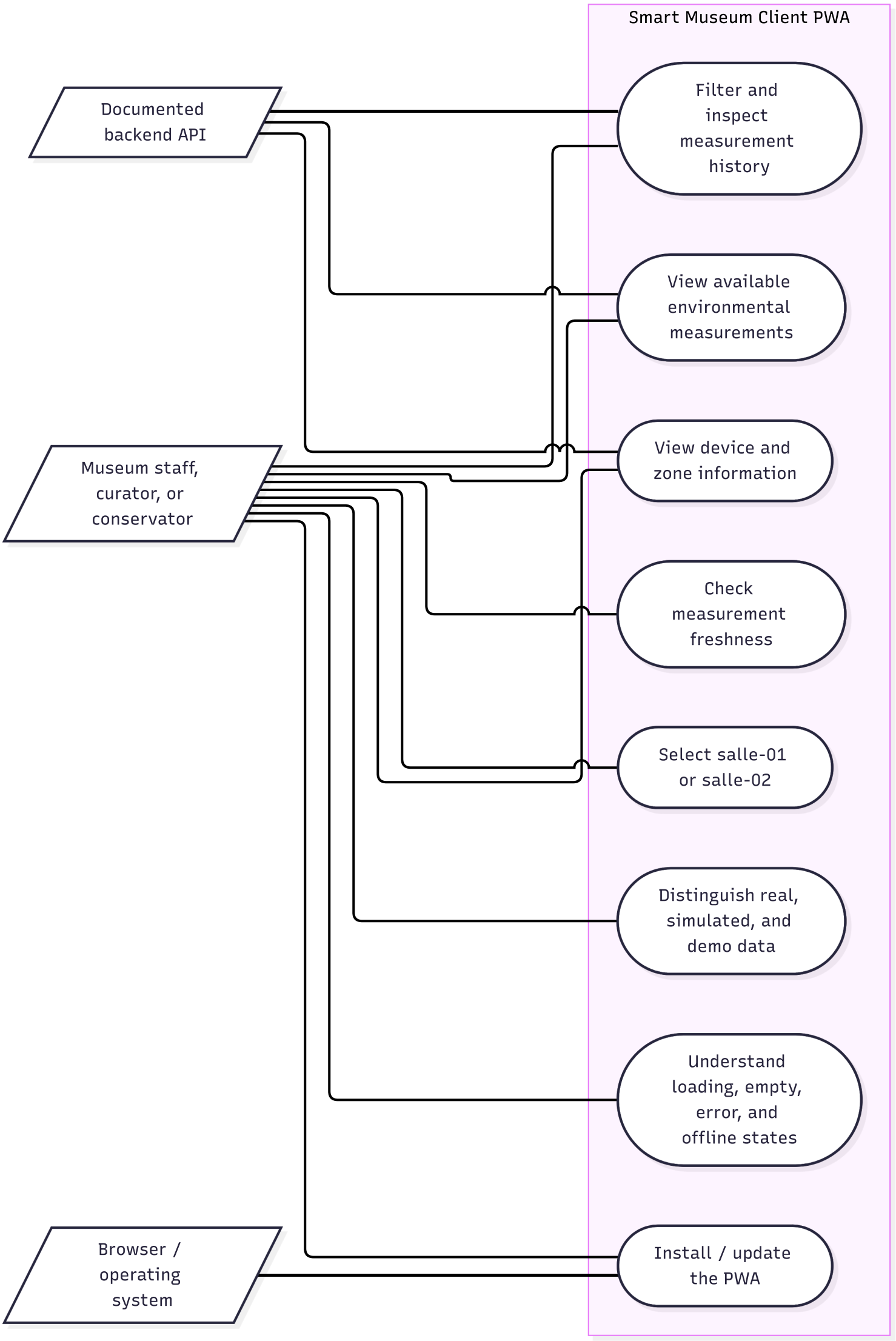
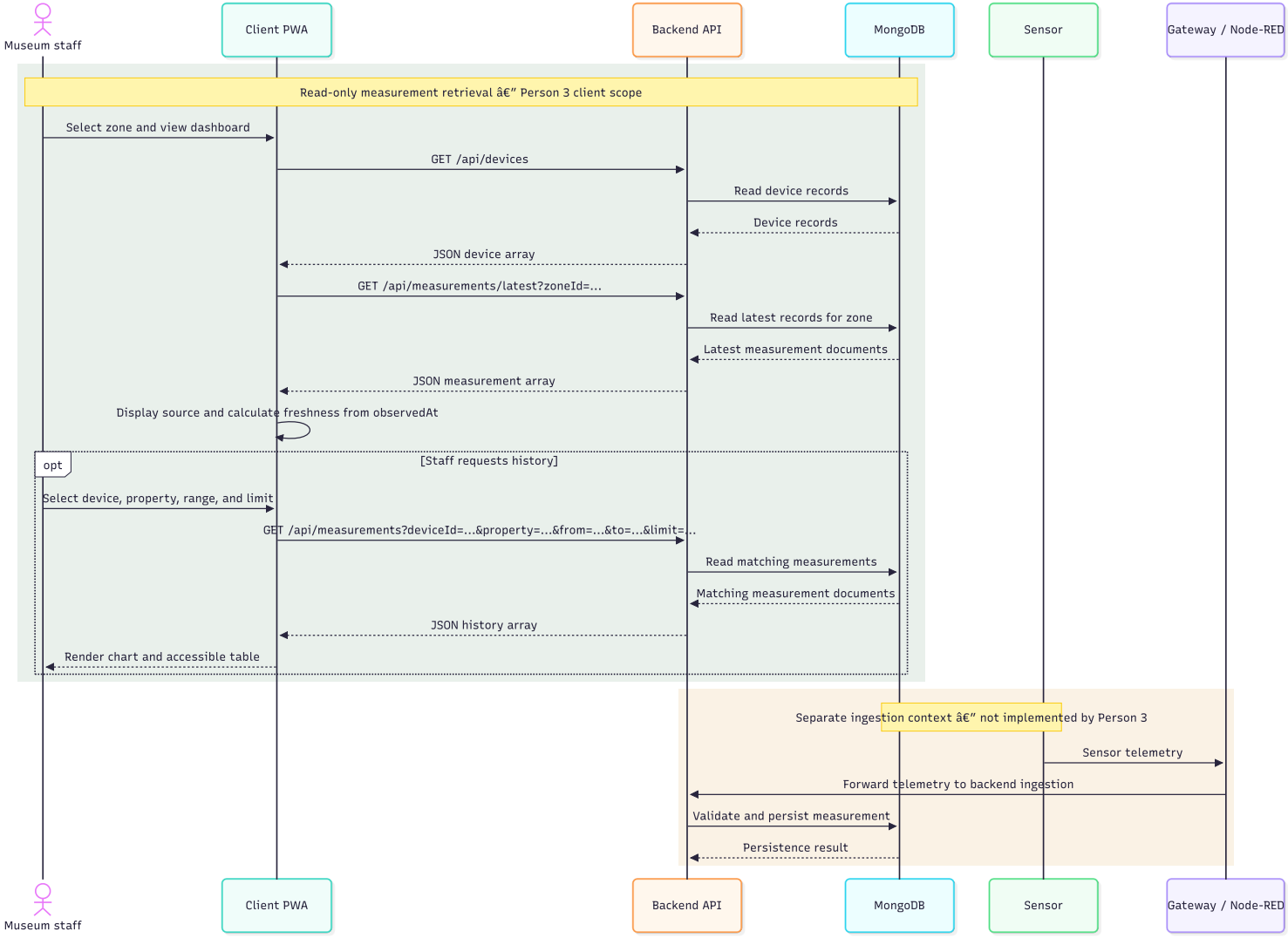
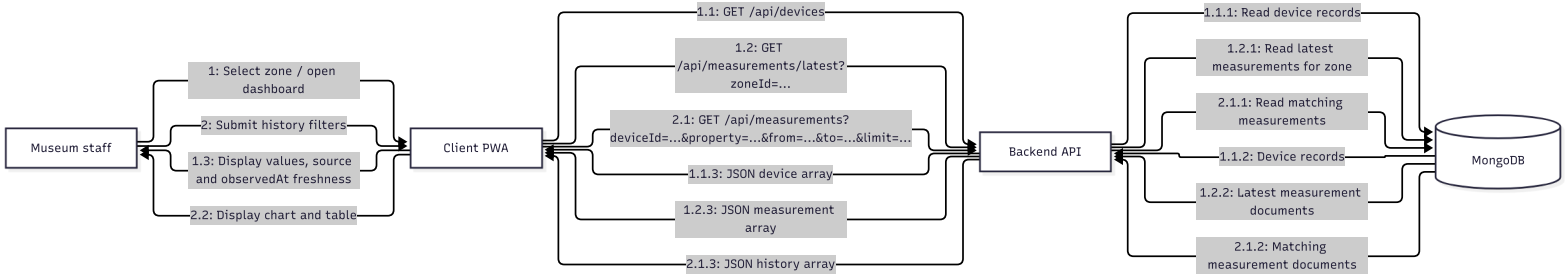

Smart Museum Client PWA
Documented backend API
Museum staff, curator, or conservator
Browser / operating system
Filter and inspect measurement history
View available environmental measurements
View device and zone information
Check measurement freshness
Select salle-01 or salle-02
Distinguish real, simulated, and demo data
Understand loading, empty, error, and offline states
Install / update the PWA


Museum staff
Client PWA
Backend API
MongoDB
Sensor
Gateway / Node-RED
Read-only measurement retrieval â€” Person 3 client scope
Select zone and view dashboard
GET /api/devices
Read device records
Device records
JSON device array
GET /api/measurements/latest?zoneId=...
Read latest records for zone
Latest measurement documents
JSON measurement array
Display source and calculate freshness from observedAt
opt
[Staff requests history]
Select device, property, range, and limit
GET /api/measurements?deviceId=...&property=...&from=...&to=...&limit=...
Read matching measurements
Matching measurement documents
JSON history array
Render chart and accessible table
Separate ingestion context â€” not implemented by Person 3
Sensor telemetry
Forward telemetry to backend ingestion
Validate and persist measurement
Persistence result
Museum staff
Client PWA
Backend API
MongoDB
Sensor
Gateway / Node-RED


Museum staff
1: Select zone / open dashboard
2: Submit history filters
1.3: Display values, source and observedAt freshness
2.2: Display chart and table
Client PWA
1.1: GET /api/devices
1.2: GET /api/measurements/latest?zoneId=...
2.1: GET /api/measurements?deviceId=...&property=...&from=...&to=...&limit=...
1.1.3: JSON device array
1.2.3: JSON measurement array
2.1.3: JSON history array
Backend API
1.1.1: Read device records
1.2.1: Read latest measurements for zone
2.1.1: Read matching measurements
1.1.2: Device records
1.2.2: Latest measurement documents
2.1.2: Matching measurement documents
MongoDB